

Sophia Li

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PROFILE

Mechanical Engineering and Aeronautics & Astronautics student at Stanford University with hands-on, full-cycle experience in aerospace structures and mechanical design, spanning CAD modeling, structural/FEA analysis, composite manufacturing, and physical testing. Skilled at translating structural requirements into manufacturable, test-validated hardware. Passionate about applying rigorous structural analysis and hands-on fabrication skills to ensure the safety and reliability of aerospace systems.

EDUCATION

Mechanical Engineering, Stanford University

Palo Alto, CA | **09/2025 – 06/2029**

GPA: 4.048

Relevant Coursework: CS223A/ME320 – Introduction to Robotics, AA100 – Introduction to Aeronautics and Astronautics, CME102 – Ordinary Differential Equations for Engineers, CS109 – Probability for Computer Scientists, CS106A/B – Programming (Python, C++)

SKILLS

- **Hardware & Manufacturing:** Flight Hardware Development, Rapid Prototyping, Design for Manufacturing (DFM), Composites Manufacturing (wet layup), 3D Printing, Fabrication Tooling, Soldering, Circuit Wiring, Woodworking
- **CAD & FEA:** Autodesk Fusion 360, SolidWorks, Abaqus (FEA), GD&T, HyperX (Collier Aerospace)
- **Structural Analysis:** Classical Laminate Theory, Composite Structures, Mechanics of Materials, Failure Criteria
- **Aerodynamics:** XFLR5, Airfoil Analysis, Lift Distribution Modeling, Wing Sizing
- **Programming:** Python, C++, MATLAB

ENGINEERING PROJECTS

Composite Wing Design, Analysis, Manufacturing, & Testing

06/2026 – 07/2026

- Designed and manufactured a carbon fiber composite foam-core NACA 2412 wing in Fusion 360, carrying the full structural design cycle from aerodynamic loading through physical prototype and test.
- Fabricated flight-representative composite wing hardware using wet layup and foam-core manufacturing techniques and built 2D/3D Abaqus composite shell models validating structural response within 2% of analytical predictions.
- Performed structural failure analysis using HyperX Tsai-Wu criteria ($MS = 76.5$) and buckling analysis ($MS = 26.7$) to evaluate design margins, the same margin-of-safety approach used in assessing composite structural repairs and damage tolerance.
- Developed MATLAB-based tools to evaluate laminate stiffness, neutral axis location, and bending response, informing layup and spar design decisions.
- Physically validated the composite wing panel through cantilever whiffletree load testing at full design load, confirming elastic structural survival with no permanent deformation

Custom Hot Wire Cutter Design & Fabrication

07/2026

- Independently proposed and constructed a nichrome-wire hot wire cutter with a tensioning mechanism for precision, straight-line foam-core cuts, applying electrical, thermal, and mechanical design knowledge to enable rapid prototyping of aerodynamic foam cores.

EXPERIENCE

Mechanical Design Lead, Stanford Design/Build/Fly

Stanford University | **09/2025 – Present**

- Leads mechanical design and prototyping for an AIAA competition aircraft, managing 10 engineers across design, manufacturing, and testing.
- Designed, fabricated, and tested a two-stage servo-actuated mechanical release mechanism, a compact electromechanical assembly integrating servos with flight hardware requirements.
- Translated aircraft performance requirements into manufacturable mechanical subsystem designs, managing full aircraft CAD integration in Fusion 360 and created manufacturing documentation to accelerate prototype development and fabrication cycles.

Research Assistant, Stanford Freeman Spogli Institute (FSI)

Stanford University | **04/2026 – Present**

- Sole author of the lab's IRB application; facilitated technical communication between faculty leads and executed quantitative analysis of large-scale survey data, developing visualizations to communicate findings to collaborators.

VOLUNTEER EXPERIENCE

Outreach, Stanford STEM Nights

Stanford University | **10/2025 – Present**

- Coordinated cross-functional teams to deliver technical demonstrations for large-scale STEM engagement events, ensuring safety compliance and effective public communication.

AWARDS

- 2024 + 2025 Morley Award Recipient (~23/1161) — awarded to Georgia Tech's distance math students whose final grades in linear algebra and multivariable calculus ranked in the top 2% of the class.